

SIDHANT BANSAL

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience in full-stack delivery and applied data engineering. At Washington State, build Python simulation and risk models analyzing 10M+ AIS vessel-tracking messages and maintain a statewide C#/ASP.NET incident-management platform, while integrating AI-assisted development tools across both environments.

WORK EXPERIENCE

Washington State Department of Ecology

Software Engineer

Seattle, WA

Aug 2023 – Present

- Built a Python ETL pipeline and risk model evaluating tug escort effectiveness against oil spill and vessel casualty risk, processing 10M+ AIS vessel-tracking messages spanning 2015–2025 and visualizing results in ArcGIS for state stakeholders.
- Cut ingestion time for 1M+ row weather datasets from hours to minutes by migrating the pipeline from Pandas to Polars, and cut database write time 5x by migrating geometry storage from WKT to WKB with SQLAlchemy chunking.
- Built classification algorithms surfacing 50+ escort events annually across 2.5M+ AIS records (3 tug and 5 tanker categories), translating speed, proximity, and duration thresholds into code.
- Reduced vessel drift simulation runtime from 1 hour to 5 minutes by optimizing the momentum and drift module with Numba and Polars, modeling wind and current effects on vessels after loss of propulsion.
- Restructured 500M+ annual weather records into an in-memory 3D matrix, cutting per-step lookup time from ~40ms to ~2µs across 3000 simulation steps per vessel.
- Maintain the Spills Program's statewide incident response platform on a 3 developer team, translating requirements from 5+ stakeholder sections into shipped tickets.
- Decreased incident logging time 40% by redesigning the platform's database layer restructuring a one to many relationship and adding new tables and stored procedures.
- Facilitate monthly alignment sessions with 5+ sections of the Spills Program to surface operational pain points and scope fixes, shipping updates via ASP.NET/C#, SQL Server, Azure DevOps, and GitHub Copilot.

Platform Science

Software Developer Intern

San Diego, CA

Jun 2022 – Sep 2022

- Improved client side integration productivity by 20% and development velocity by 15% by shipping backward compatible RESTful APIs in Node.js, Express, and Laravel (PHP), refactoring JS/TypeScript and PHP codebases, and introducing automated unit tests via Docker based CI.

Accenture

Application Developer

Bangalore, India

Sep 2019 – Aug 2021

- Lowered API response time 40%, supported 30% more concurrent users through caching and database indexing, and lifted user engagement 25% through UI/UX and API optimizations, building and scaling React.js/MEAN-stack applications and Node.js/Express backends in partnership with PMs within Agile sprints.

TECHNICAL SKILLS

Languages: Python, C#, JavaScript, TypeScript, SQL, Java, Angular, C++

Frameworks & Cloud: React, Node.js, Express, ASP.NET, Spring Boot, AWS, Azure, Docker, ArcGIS, Unity

Databases & Tools: PostgreSQL, MySQL, MongoDB, SQL Server, Git, JIRA, GitHub Copilot, Numba, Polars, SQLAlchemy

KEY PROJECTS

VR Therapy Platform

Sep 2022 - Jun 2023

- Unified four separate VR therapy applications into a single, modular Unity/C# platform, reducing code redundancy by 35% and improving rendering performance by 20% through optimized draw calls.

Dependify

Apr 2022 – Jun 2022

- Dropped response time 40% at 10,000+ daily requests by architecting a distributed microservices system (Java, Spring Boot, PostgreSQL, AWS EC2) for centralized dependency management.

EDUCATION

University of Washington

Master of Science, Computer Science

Seattle, WA

Sep 2021 – Jun 2023

Guru Gobind Singh Indraprastha University

Bachelor of Technology, Information Technology

New Delhi, India

Aug 2015 – Jun 2019